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Influence of the wear resistance of the working tool material on hydraulic breaker productivity

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Abstract

During operation of hydraulic breakers, the dimensions of the blunting area of the pick tip increase owing to impact-abrasive wear. This leads to an increase in the contact area of the tool with the rock on impact, with a reduction in the intensity of impact fracture. As a result, from a certain moment onward the process of tool penetration into the rock practically ceases, and to resume it the pick has to be replaced or resharpened. Given that these operations are very labour-intensive, the selection of wear-resistant materials for manufacturing the working tools of hydraulic breakers and the development of methods that increase their resistance to impact-abrasive wear and, consequently, the service time of picks before resharpening are an urgent task.

Using the JCB HM380 hydraulic breaker (impact energy A_0 = 981 J) as an example, the influence of the wear rate of its working tool (a cylindrical pick with ends in the form of a flat face and a conical tip with a spherical blunting of radius $R_п$ that changes as the pick tip wears; made of two steels used for such purposes and differing in hardness, 38XM and X12МФА$_х$ after cold treatment) on the productivity of the machine when breaking a rock massif ($\sigma_{сж}$ = 50 MPa) was analysed.

Using the developed mathematical model of impact rock fracture, the penetration depth of the pick into the rock per impact $\alpha_{внi}$ and the total penetration $\Sigma\alpha_{вн}$ over a cycle of impacts (Fig. 1) were calculated and compared with the results of cinephotographic measurements of pick penetration into concrete during operation of the JCB HM380 hydraulic breaker (Fig. 2). Satisfactory agreement between the calculated and experimental results was noted.

[Figure 1 — see original PDF, page 1] Calculated dependence of the total penetration depth of the pick into the rock on the number of impacts, compared with the results of cinephotography (♦)

[Figure 2 — see original PDF, page 1] Cinephotogram of the pick penetration process into rock (n = 1, n = 30), used to establish the penetration depth

[Figure 3 — see original PDF, page 1] Change in the blunting radius of the pick tip (●) and of the total penetration depth $\Sigma\alpha_{вн}$ into the rock (■) with the number of impacts for picks made of 38XM steel (solid line) and X12МФА$_х$ steel (dashed line)

It is shown that the use of the wear-resistant steel X12МФА$_х$ as the pick material can increase by a factor of 2.8 (from n^*_1 = 29272 to n^*_2 = 80844) the number of impacts delivered by the pick until it is completely blunted, and the total depth of penetration into the rock (from 12.5 m to 34.6 m),

compared with a pick made of a less wear-resistant material, 38XM steel, thereby substantially increasing the productivity of the hydraulic breaker and extending the service life of its working tool before replacement or resharpening.